

Name: \_\_\_\_\_

Seat #: \_\_\_\_\_

## FINAL EXAMINATION

ECON 201, Spring 2014

Prof. Ryan Edwards

May 19, 2014

11am–1pm

Kiely 170

### **RULES:**

- Don't start until I say so. **DON'T LOOK THROUGH THE EXAM.**
- But please check that you have all 9 pages.
- Write your name and **seat number** at the top.
- When I call time, **stop**. If you don't, you get an F.
- No books, no notes, no calculators.
- There are 3 questions worth a total of 90 possible points, one for each minute of time you have. Budget your time accordingly.
- If I catch you cheating in any way, you will receive an F for the course and a referral to the VP for Student Affairs.

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|---------|---|---|---|---|---|--------|
| Points: | 1 | 2 | 3 | 4 | 5 | Total: |
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1. (20 points total) *Short answer questions.*

(a) (5 points) Recall that  $MV = PY$ . What does this relationship tell you about the long-run rate of inflation if the long-run rate of money growth is 5%, the long-run rate of income growth is 3%, and velocity is not changing? Be specific and show your work.

(b) (5 points) Recall the Fisher equation, relating the nominal interest rate  $i$  to the real interest rate  $R$  and inflation:  $i = R + \pi$ . What is the lowest real interest rate  $R$  the Fed can obtain if  $\pi = -0.01$  and why? Do you think this might be beneficial or problematic? Discuss.

Remember that national income accounting suggests that  $S - I = NX$ , where  $S$  is domestic saving,  $I$  is domestic investment, and  $NX$  is the trade balance. Further,  $S = [Y - T - C] + [T - G]$  where  $T$  is taxes. Finally, assume  $NX$  is constant.

(c) (5 points) If the government cuts taxes,  $T$ , would that help or hurt investment,  $I$ , or would it do nothing at all? Explain.

(d) (5 points) If the government cuts spending,  $G$ , would that help or hurt investment,  $I$ , or would it do nothing at all? Explain.

2. (34 points total) *How Solow can you go?*

Imagine the U.S. is a Solow economy in which the saving rate is  $\bar{s}$ , depreciation is  $\bar{d}$ , capital is  $K$ , population is  $L$ , total factor productivity is  $\bar{A}$ , and GDP is

$$Y = F(\bar{A}, K, L) = \bar{A}K^{1/3}L^{2/3}. \quad (1)$$

Capital is accumulated according to the standard flow equation, where the change in the capital stock is equal to saving minus depreciation,

$$\Delta K = \bar{s}Y - \bar{d}K. \quad (2)$$

Finally, the wage equals the marginal product of labor, given by

$$w = MPL = \frac{2}{3}\bar{A}\left(\frac{K}{L}\right)^{1/3} \quad (3)$$

- (a) (2 points) Recall the definition of the steady state. What does it imply about equation (2)? Write it down and describe.

- (b) (5 points) Using your answer to part (a), substitute for  $Y$  using equation (1), and solve for  $K^*$ , the level of capital in the steady state, as a function of  $s$ ,  $d$ ,  $A$ , and  $L$ .

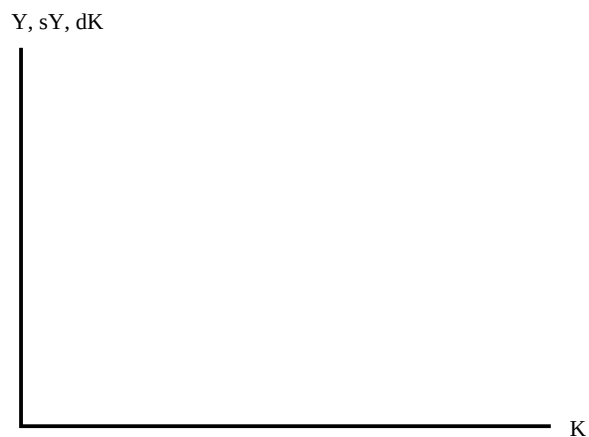
- (c) (3 points) Take the growth rate of both sides of your answer to part (b) and simplify.

What determines the saving rate,  $s$ ? A bunch of things, probably; family size and composition matter, and so does education and probably wealth. What about *taxes on capital*, which raise the “price” of saving? (When capital taxes are higher, the price of saving is higher.)

If we were in Econ 202 right now, we’d have to worry about income versus substitution effects. But in Econ 201, we typically assume that substitution effects dominate, and we’ll do so here too. That means that when the price of something rises, we’ll assume that people substitute away from it.

- (d) (3 points) Using this logic, would the saving rate rise, fall, or stay the same if the tax rate on capital was higher? Explain fully.

- (e) (4 points) Illustrate what you have described in part (d) using the Solow diagram below. Show the effects, if any, of an increase in the tax on capital. Explain fully.



We could also tax labor instead of capital. A tax on labor reduces the price of leisure and effectively raises the price of labor.

- (f) (3 points) Using this logic, would labor supplied rise, fall, or stay the same if the tax rate on labor was higher? Explain fully.

If we can raise revenue either by taxing labor or capital, how do we decide which one is better to tax? One perspective is to compare reductions in the steady-state capital stock  $K^*$  that result.

(g) (4 points) Using your answer to part (c), identify which reduces steady-state  $K^*$  more: a 1% reduction in the saving rate,  $s$ , or a 1% reduction in labor supplied,  $L$ . Explain.

(h) (5 points) Suppose we knew that labor supply was much less responsive to taxes, or *more inelastic*, than saving. In which way would this tilt the decision, toward taxing labor or taxing capital, or would it leave the decision unaffected? Explain.

(i) (5 points) Suppose we moved from theory to real-world analysis, and in so doing we recognized that many workers do not own capital at all, and thus are relatively poor, but they do supply labor. Meanwhile, those who own capital tend to be considerably more wealthy than those who do not. In which way would this tilt the decision, toward taxing labor or taxing capital, or would it leave the decision unaffected? Explain.

3. (36 points total) *Thanks a lot, Generation X, You Broke the Short Run.*

“Generation X” is the term for Americans born roughly between the 1960s and 1980s. People born later are commonly called “Millennials.” Recall the standard IS, MP, and PC Curves we saw in class:

$$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r}) \quad (4)$$

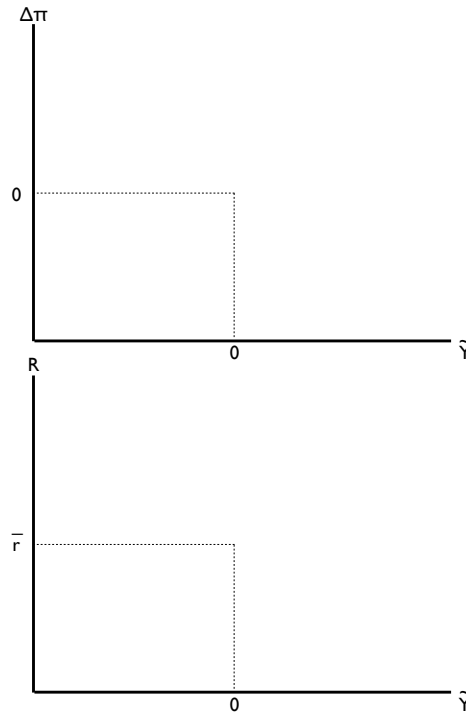
$$R = R^{ff} + \bar{p} \quad (5)$$

$$\Delta\pi = \bar{v}\tilde{Y} + \bar{o} \quad (6)$$

where  $R$  is the real interest rate paid by borrowers;  $\bar{r}$  is the marginal product of capital;  $R^{ff}$  is the real federal funds rate;  $\bar{p}$  is the risk premium charged by private lenders;  $\pi$  is the inflation rate; and  $\bar{o}$  is an external inflation shock.

Imagine that it is 2002, and the economy has emerged from the recession of 2001. Generation X’ers feel optimistic about the future, and they increase spending on housing as a share of permanent income, racking up debt.

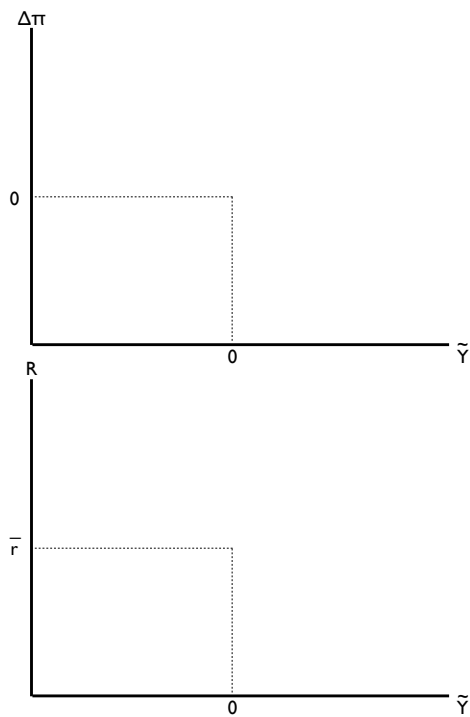
- (a) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you’ve drawn. *If a curve shifts twice in this answer, show that.*



Now imagine that it is 2006, and a major crisis hits the housing market. Generation X'ers discover that their incomes are not sufficient to make their mortgage payments on the enormous houses they starting buying in 2002.

Homeowners sharply reduce their consumption spending in hopes of avoiding default on their mortgages. But it's too little, too late.

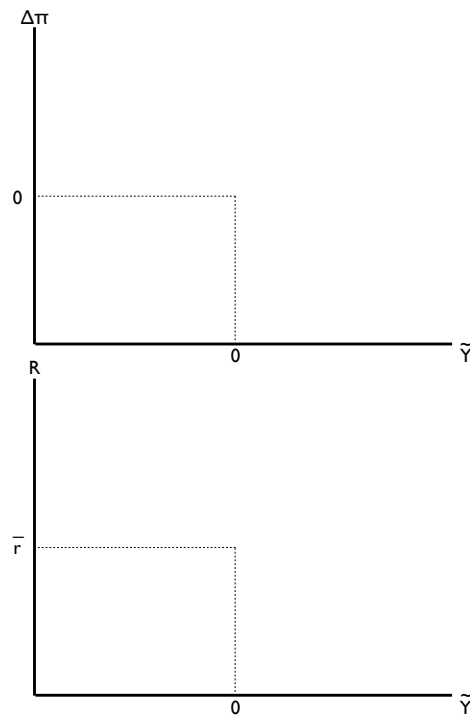
- (b) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. *If a curve shifts twice in this answer, show that.* [You don't need to show the events from part (a).]



Despite their efforts, many Generation X'ers default on their mortgages, and lenders lose their loaned funds. Unfortunately, those “lenders” aren’t just a few banks. Because the original mortgage lenders sold those mortgage agreements off to many different banks and investors, many agents in the economy lose assets. Even worse, remaining lenders don’t know exactly which agents lost a lot in the mortgage market, only that many of them did and thus are not good borrowers because they’re near bankruptcy.

- (c) (6 points) If you were a good lender during this period, with funds available, and a borrower walked into your office and asked to borrow money, what interest rate would you charge him or her? Would it be higher, lower, or the same as the usual interest rate you’d charge, and why?

- (d) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you’ve drawn. *If a curve shifts twice in this answer, show that.* [You don’t need to show the events from parts (a) or (b).]





- (e) (6 points) If the Federal Reserve had already reduced the nominal federal funds rate to zero before the events of part (d), would that change your story at all, and if so how? Explain.
- (f) (6 points) What else could the government use to address the issues afflicting the economy in part (d) other than the Federal Reserve's monetary policy? Illustrate this alternative in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. [You *should* show the events from part (d) but not from parts (a) or (b).]

